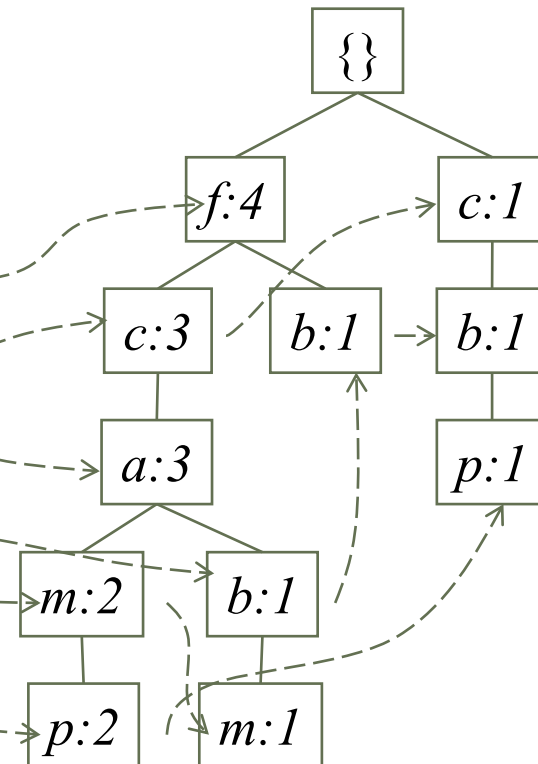


Mining FP-Tree: Divide and Conquer Based on Patterns and Data

- Pattern mining can be partitioned according to current patterns
 - Patterns containing p : p 's conditional database: $fcam:2, cb:1$
 - p 's conditional database (i.e., the database under the condition that p exists):
 - *transformed prefix paths* of item p
 - Patterns having m but no p : m 's conditional database: $fca:2, fcab:1$
 -

min_support = 3

Item	Frequency	Header
f	4	
c	4	
a	3	
b	3	
m	3	
p	3	



Conditional database of each pattern

<u>Item</u>	<u>Conditional database</u>
<i>c</i>	<i>f:3</i>
<i>a</i>	<i>fc:3</i>
<i>b</i>	<i>fca:1, f:1, c:1</i>
<i>m</i>	<i>fca:2, fcab:1</i>
<i>p</i>	<i>fcam:2, cb:1</i>

Mine Each Conditional Database Recursively

min_support = 3

Conditional Data Bases

<i>item</i>	<i>cond. data base</i>
<i>c</i>	<i>f:3</i>
<i>a</i>	<i>fc:3</i>
<i>b</i>	<i>fca:1, f:1, c:1</i>
<i>m</i>	<i>fca:2, fcab:1</i>
<i>p</i>	<i>fcam:2, cb:1</i>

□ For each conditional database

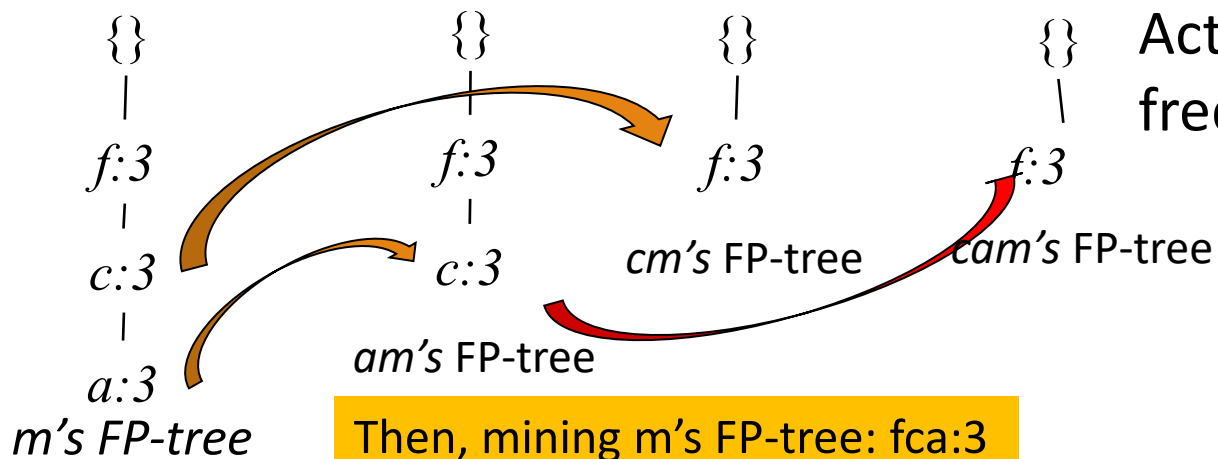
□ Mine single-item patterns

□ Construct its FP-tree & mine it

p's conditional DB: *fcam:2, cb:1* → *c: 3*

m's conditional DB: *fca:2, fcab:1* → *fca: 3*

b's conditional DB: *fca:1, f:1, c:1* → ϕ



Actually, for single branch FP-tree, all the frequent patterns can be generated in one shot

m: 3

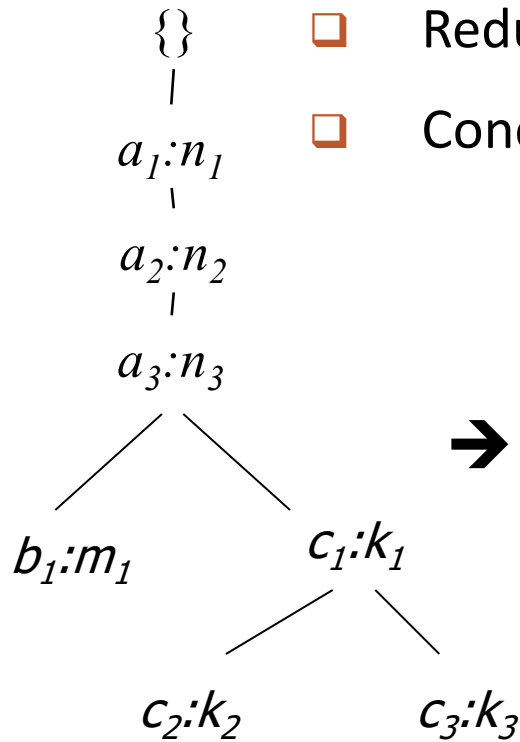
fm: 3, cm: 3, am: 3

fcm: 3, fam:3, cam: 3

fcam: 3

A Special Case: Single Prefix Path in FP-tree

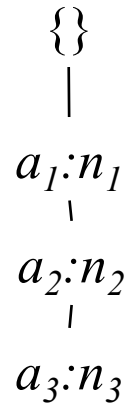
- Suppose a (conditional) FP-tree T has a shared single prefix-path P
- Mining can be decomposed into two parts



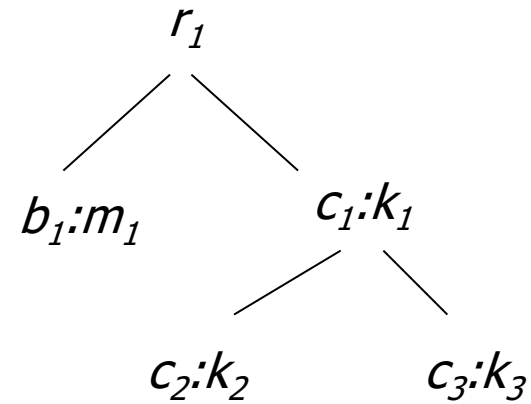
- Reduction of the single prefix path into one node
- Concatenation of the mining results of the two parts



$r_1 =$



+



FPGrowth: Mining Frequent Patterns by Pattern Growth

- ❑ Essence of frequent pattern growth (FPGrowth) methodology
 - ❑ Find frequent single items and partition the database based on each such single item pattern
 - ❑ Recursively grow frequent patterns by doing the above for each *partitioned database* (also called the pattern's *conditional database*)
 - ❑ To facilitate efficient processing, an efficient data structure, FP-tree, can be constructed
- ❑ Mining becomes
 - ❑ Recursively construct and mine (conditional) FP-trees
 - ❑ Until the resulting FP-tree is empty, or until it contains only one path—single path will generate all the combinations of its sub-paths, each of which is a frequent pattern